

```
1 using System;
2 using System.Collections.Generic;
3 using System.Linq;
4 using System.Text;
5 using System.Text.Json;
6 using System.Threading.Tasks;
7 using CustomProjectRPG.Interfaces;
8
9 namespace CustomProjectRPG
10 {
11     {
12         public class APComponent: ISerializeJSON<APComponent>
13         {
14             private int _currentAP;
15             private const int _maxAP = 5;
16
17             public APComponent()
18             {
19                 _currentAP = _maxAP;
20             }
21
22             public int CurrentAP
23             {
24                 get { return _currentAP; }
25             }
26
27             public int MaxAP
28             {
29                 get { return _maxAP; }
30             }
31
32             public bool UseAP(int amount)
33             {
34                 if (amount <= 0)
35                 {
36                     throw new ArgumentException("AP usage must be greater than
37                                     zero.");
38                 }
39
40                 if (_currentAP < amount)
41                 {
42                     return false;
43                 }
44
45                 _currentAP -= amount;
46                 return true;
47             }
48
49             public void Reset()
```

```
49     {
50         _currentAP = _maxAP;
51     }
52
53     public bool IsDepleted()
54     {
55         return _currentAP == 0;
56     }
57
58
59     public string Serialize()
60     {
61         return JsonSerializer.Serialize(_currentAP);
62     }
63
64     public void Deserialize(string json)
65     {
66         if (string.IsNullOrEmpty(json)) return;
67         int value = JsonSerializer.Deserialize<int>(json);
68         _currentAP = Math.Clamp(value, 0, _maxAP);
69     }
70
71 }
72
73
74 }
75
```